

A Marketplace Model of College Tennis Recruiting: Forecasting Post-Junior Progression from Junior Metrics and Collegiate Pathways

Introduction:

College tennis recruiting is a two-sided decision: coaches must identify junior players with the greatest long-term potential, while players must choose collegiate pathways that best support their progression. Yet these decisions rely heavily on current ratings, rankings, and perceptions of college programs, with limited evidence regarding which factors actually predict future performance. I developed dual-perspective marketplace model to answer two practical questions: what information available during recruitment best predicts a junior player's future level, and does subsequent collegiate pathway provide additional predictive information?

Methods:

I retrospectively followed players from the 2022 USTA Boys' 18s National Championships ("Kalamazoo") through May 2026, using May 2026 Universal Tennis Rating (UTR) as the outcome. Player data were compiled from USTA, UTR, World Tennis Number (WTN), TennisRecruiting.net (TRN), and college athletics websites. Candidate recruiting-time predictors included pre- and post-tournament ratings, USTA and TRN rankings, high-school ranking trajectories, graduation cohort, and geographic region. Univariable and multivariable linear regression were used for model development. The Coach's Recruiting Model incorporated post-tournament UTR (PT UTR), senior-year TRN ranking (TRN12), proportional ranking trajectory (TRN12/TRN9), graduation cohort, and region. The Player's Career Model incorporated PT UTR, TRN12, proportional ranking trajectory (TRN12/TRN9), and May 2026 college/career pathway. Incremental contributions were assessed using nested-model F-tests and change in R^2 ; model fit and out-of-sample prediction were evaluated using adjusted R^2 and 5-fold cross-validated root mean squared error (CV-RMSE), respectively.

Results:

PT UTR was the strongest individual recruiting-time predictor of May 2026 UTR ($R^2=.635$), with pre-tournament UTR nearly equivalent ($R^2=.632$); TRN12 also strongly predicted future level ($R^2=.520$). The Coach's Recruiting Model ($n=213$) explained 68.5% of future UTR variance (adjusted $R^2=.671$) and achieved CV-RMSE=.696. The Player's Career Model explained 70.7% of variance (adjusted $R^2=.694$) with lower out-of-sample prediction error (CV-RMSE=.675), approximately 3% lower than the Coach's Model. College/career pathway collectively added 5.4 percentage points of explained variance beyond PT UTR, TRN12, and proportional trajectory ($\Delta R^2=.054$; $p<.001$). Relative to Other D1 players, adjusted future UTR was 0.13 points higher for Power-4 and 0.04 lower for Ivy players, neither statistically significant; D3 players were 0.63 points lower ($p<.001$), while professionals were 0.97 points higher ($p=.005$). Power-4 affiliation therefore provided no independent predictive advantage after accounting for junior performance characteristics.

Conclusion:

Long-term tennis performance was predicted not only by current junior ability, but also by recruiting rank, developmental trajectory, and subsequent competitive pathway. The Coach's Recruiting Model provides programs a quantitative framework for evaluating recruits beyond snapshot ratings, while the Player's Career Model adds pathway information and modestly improves out-of-sample prediction. These stakeholder-specific analytics can support both sides of the college-tennis marketplace while distinguishing meaningful predictive information from prominent but non-predictive labels such as Power-4 affiliation.